Exp: 4 Implement packet sniffing using raw pockets in python 30/07/24

AIM:

To implement a packet sniffer using raw sockets in python that captures live network traffic & parses ethernet IP & transport headers.

PROCEDURE:

Step 1: Set up Environment & confirm python 3 is installed.

Step 2: Understand packet structure you'll parse

Step 3: Create raw socket, create a file sniffer.py & start

Step 4: parse ethernet header, add function to unpack first 14 bytes

Step 5: parse the IP header

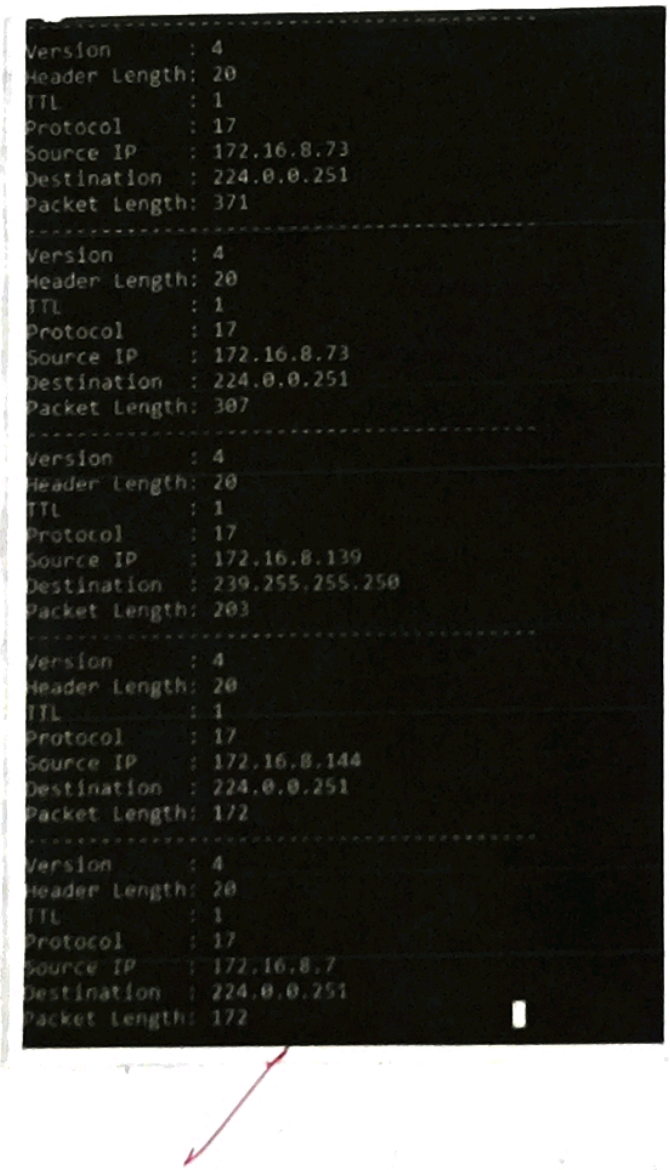
Step 6: parse TCP / UDP / ICMP headers

Step 7: Write main sniffing loop

Step 8: Run the sniffer

Step 9: Stop the sniffer with Ctrl + C


import socket
u1.py
C: > Users > HOC0222008 > Downloads > Packet Sniffer > u1.py > ...
import socket
import struct
sniffer = socket.socket(socket.AF_INET, socket.SOCK_RAW, socket.IPPROTO_IP)
host = socket.gethostbyname(socket.gethostname())
sniffer.bind((host, 0))
sniffer.setsockopt(socket.IPPROTO_IP, socket.IP_HDRINCL, 1)
sniffer.ioctl(socket.SIO_RCVALL, socket.RCVALL_ON)
print("Packet Sniffer Started...\n")
try:
while True:
raw_data, addr = sniffer.recvfrom(65535)
ip_header = raw_data[0:20]
iph = struct.unpack('!BBHHBBH4s4s', ip_header)
version = iph[0] >> 4
ihl = iph[0] & 0xF
ttl = iph[5]
protocol = iph[6]
source = socket.inet_ntoa(iph[8])
destination = socket.inet_ntoa(iph[9])
print("----------------------------------------")
print("Version :", version)
print("Header Length:", ihl * 4)
print("TTL :", ttl)
print("Protocol :", protocol)
print("Source IP :", source)
print("Destination :", destination)
print("Packet Length:", len(raw_data))
except KeyboardInterrupt:
sniffer.ioctl(socket.SIO_RCVALL, socket.RCVALL_OFF)
print("\nStopped")


Version : 4
Header Length: 20
TTL : 1
Protocol : 17
Source IP : 172.16.8.73
Destination : 224.0.0.251
Packet Length: 371
Version : 4
Header Length: 20
TTL : 1
Protocol : 17
Source IP : 172.16.8.73
Destination : 224.0.0.251
Packet Length: 307
Version : 4
Header Length: 20
TTL : 1
Protocol : 17
Source IP : 172.16.8.139
Destination : 239.255.255.250
Packet Length: 203
Version : 4
Header Length: 20
TTL : 1
Protocol : 17
Source IP : 172.16.8.144
Destination : 224.0.0.251
Packet Length: 172
Version : 4
Header Length: 20
TTL : 1
Protocol : 17
Source IP : 172.16.8.7
Destination : 224.0.0.251
Packet Length: 172

RESULT :

Hence, the Implementation of packet Sniffer has been verified & successful.